SUPPLEMENTARY MATERIAL: GOOD SPLITS ARE HARD TO FIND: UNDERSTANDING PERFORMANCE–ENERGY TRADE-OFFS IN HETEROGENEOUS EDGE INFERENCE

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A SCOPE AND QUALITY CHECKS

The deployment comparison contains 204 Native splits, their 204 Generic counterparts, and 42 Full alternatives across seven models and three setups. All 450 configurations have throughput, energy, and numerical dataset-quality observations (Mika, 2026). The broader compiler-feasibility study has 420 candidates and is separate from this paired comparison.

The numerical filter uses $(q_{\text{ref}} - q)/q_{\text{ref}} \leq \epsilon$, with CPU ONNX as the reference. At $\epsilon = 0.05$, it retains 187 Native splits, 187 Generic splits, and 33 Full alternatives. The 43 excluded configurations comprise 17 Native splits, nine accelerator Fulls, and 17 Generic splits. Both runners retain exactly the same cuts. All seven global split winners and their Full references remain unchanged.

Table 1 applies each threshold to splits and Full alternatives before reselecting the local Full references. Local optima can change: MobileNet/H10 b078 loses 8.96% relative Top1 accuracy. At 5%, b074 becomes the admissible optimum at 1,142.90 FPS. Generic still selects b027, whose Native throughput loss decreases from 27.58% to 25.75%.

For 53 Native DeepX classification records, cached loss labels disagree with the numerical evaluation. We use the numerical values and retain both sources. Eligibility is retrospective and assumes that these quality measurements are already available.

The 5,000-example evaluations use 1,000 paired-bootstrap replicates. Their intervals compare each deployment with its CPU reference, not a split directly with a Full. The retained YOLOv7 summaries contain no per-image predictions or bootstrap samples for a direct split-minus-Full interval. Runtime parity checks on prepared inputs are separate from dataset-quality evaluation. The artifact bindings are documented in Section D.

B FULL REFERENCES AND MEASUREMENT REPEATS

Selection uses the highest eligible FPS median for each model, with configuration identity breaking exact ties. All seven selected splits use Native. Table 2 keeps each selected split fixed and gives its same-host and global Full references. Table 3 lists all 42 host-specific Full alternatives and their quality values. The TensorRT averages in the main paper summarize hosts. Selection uses actual configurations.

Table 4 gives the absolute energy values behind the main paper's percentage changes. Each value belongs to the throughput-selected configuration, which need not minimize energy. Performance and energy are separate measurements of the same configuration.

Table 5 retains the observed repeat ranges for the global winners and the two additional YOLOv7 Full references. MobileNet's split and Full throughput ranges overlap, while their energy ranges remain separate. These ranges describe the recorded runs, without accounting for selection of a maximum or systematic measurement uncertainty.

C RUNTIME AND MEASUREMENT BOUNDARIES

Table 6 lists available platform and runtime records. The scope column identifies which configurations each record covers. An x86 controller coordinates the three Orin NX hosts. All 204 splits execute an accelerator prefix followed by a TensorRT suffix.

The quality records identify an ImageNet validation subset and a COCO 2017 validation set of 5,000 images each. Dataset hashes agree across all 354 classification configurations and, separately, all 96 detector configurations. These identities bind the comparisons to the same images. The compact records do not specify the subset selection algo-

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Table 1. Quality-constrained Native advantages on the same host. Each row filters both Native splits and Full alternatives using the numeric quality loss against the CPU reference, then reselects the fastest and minimum-energy Full. Faster and Lower energy compare against the same fastest Full. Both requires both advantages. Below min. Full compares energy against the lowest-energy admitted Full. These are retrospective point-estimate comparisons.

Eligibility	Native cuts	Fulls	Faster	Lower energy	Both	Below min. Full
All measured	204	42	105	77	65	72
$\leq 1\%$ loss	114	28	58	40	32	37
$\leq 2\%$ loss	147	30	88	61	52	56
$\leq 5\%$ loss	187	33	102	74	62	69

Table 2. The same split selected in the main Table 3, with two Full references. The split is not reselected by relative gain. R_h uses the fastest Full on its physical host and R_g the globally fastest Full. Consequently $R_g \leq R_h$. This comparison describes the baseline choice, not an optimization step.

Model	Selected split	FPS	Host Full	FPS	Global Full	FPS	R_h	R_g
MobileNetV3-L	DeepX/b135	1552	DeepX	1469	DeepX	1469	1.06	1.06
RegNet-X 1.6GF	H10/b123	1367	TRT	316	DeepX	538	4.33	2.54
ResNet-50	H10/b095	1100	TRT	522	DeepX	656	2.11	1.68
YOLO11l	H10/b120	53	TRT	39	DeepX	63	1.36	0.84
YOLO26m	H10/b043	110	TRT	84	TRT (H8)	85	1.31	1.30
YOLO26s	H8/b021	187	TRT	151	TRT (DeepX)	152	1.24	1.23
YOLOv7	H8/b066	113	TRT	53	DeepX	83	2.13	1.35

rithm.

Explicit preprocessing objects are retained for 298 classifier and 87 detector configurations. The classifier objects specify RGB input and direct bilinear resizing to 224×224. Detector objects specify RGB input and centered bilinear letterboxing to 640×640, with padding value 114. The remaining configurations are linked through the completed audit and source records. These contracts describe the prepared RGB uint8 image stage. They do not specify every backend-specific scaling and normalization step. The public `r20` evidence projection retains each object and source location.

The bridge inventory contains 124 split records labelled `float32_layout_fp16` and 80 labelled `uint8_dequant_fp16`. These names describe boundary representation and bridge conversion, rather than the precision of every accelerator operator. Full implementations retain their own precision contracts.

The nine pathway projections cover MobileNet/DX b001 and H8 b027, RegNet/H10 b052/b123, ResNet/H10 b002/b095, and YOLOv7/H10 b066 and H8 b044/b066. H8 b066 additionally records a postprocessing queue of four. Settings observed for these pathways are not assumed for all 204 splits.

Generic timing starts from prepared input before the first partition. Native records preserve a worker-ready start boundary. Application preprocessing precedes timing. Both paths count completed tasks, including materialization and fill/drain.

The YOLOv7 report's separate Python-loader probe records CUDART 12.9.79 on H10 and 12.6.68 on H8/DX, with driver API 12.6 on all three. This probe does not observe libraries loaded by earlier measurement processes or by the H8 C++ shared object. A separate H8 runner-import check reports NumPy 1.26.4, TensorRT 10.3 and HailoRT 4.20. It does not establish the software stack of every setup. Load samples at approximately 1 Hz cannot resolve brief intervening clock or thermal events, and the report supplies no EMC measurement.

The compact deployment inputs do not fully reconstruct checkpoint and dataset-manifest identities, exporter/opset versions, all device/compiler/runtime versions, or contemporaneous power modes, clock/thermal traces and affinity settings. The analysis environment in the reproduction metadata is distinct from device software. These missing fields limit exact reacquisition and causal attribution of cross-host differences. They do not change the recorded per-host split/Full denominators. The DeepX Native policy uses batch one, one engine per process, asynchronous submission, and at most four outstanding requests for all seven models and supported cuts. Four output slots preserve request identity and output lifetime. Splits retain an owned boundary copy, a FIFO of depth three, and a serial TensorRT consumer. Full uses the same policy without an artificial partition boundary. The policy is not selected separately for each model.

H8 YOLOv7 b066 uses three stages, whereas b009/b044 combine TensorRT and completion in one consumer. Generic and the Hailo Native paths retain their respective

Table 3. Full alternatives on all three hosts. q is Top1 accuracy for classifiers and COCO $AP_{50:95}$ for detectors, both displayed in percent.

Model	Host	Accel. FPS	mJ/image	q (%)	TRT FPS	mJ/image	q (%)
MobileNetV3-L	H8	118	126.75	60.62	749	29.16	73.64
MobileNetV3-L	H10	210	74.96	59.42	710	27.63	73.64
MobileNetV3-L	DeepX	1469	12.01	71.72	694	27.76	73.60
RegNet-X 1.6GF	H8	362	43.24	78.76	314	77.22	78.92
RegNet-X 1.6GF	H10	259	61.79	78.84	316	74.25	78.94
RegNet-X 1.6GF	DeepX	538	31.54	78.70	333	71.81	78.94
ResNet-50	H8	122	132.45	80.32	523	56.90	80.50
ResNet-50	H10	184	89.54	80.60	522	55.17	80.50
ResNet-50	DeepX	656	27.65	80.54	517	53.97	80.50
YOLO11l	H8	11	1426.62	44.15	39	680.33	46.66
YOLO11l	H10	11	1659.47	45.31	39	653.47	46.66
YOLO11l	DeepX	63	315.23	45.67	38	675.47	46.68
YOLO26m	H8	18	935.03	39.65	85	400.03	44.82
YOLO26m	H10	14	1290.53	40.55	84	390.25	44.85
YOLO26m	DeepX	80	239.32	44.27	83	389.08	44.86
YOLO26s	H8	23	701.73	32.33	151	197.98	40.09
YOLO26s	H10	15	1108.70	34.17	150	192.20	40.09
YOLO26s	DeepX	105	176.81	39.39	152	187.80	40.11
YOLOv7	H8	32	549.84	40.78	53	598.41	46.02
YOLOv7	H10	33	601.83	41.52	53	583.57	46.02
YOLOv7	DeepX	83	269.44	45.86	48	613.18	46.04

Table 4. Absolute calibrated system energy for the same global throughput-selected Full and Native split configurations as the main Table 3. These are actual configurations, not averages across hosts. Negative energy change means less split energy per image.

Model	Full	mJ/image	Split	mJ/image	Energy change (%)
MobileNetV3-L	DeepX	12.01	DeepX/b135	12.59	+4.8
RegNet-X 1.6GF	DeepX	31.54	H10/b123	15.83	-49.8
ResNet-50	DeepX	27.65	H10/b095	29.46	+6.6
YOLO11l	DeepX	315.23	H10/b120	555.75	+76.3
YOLO26m	TRT (H8)	400.03	H10/b043	319.65	-20.1
YOLO26s	TRT (DeepX)	187.80	H8/b021	157.65	-16.1
YOLOv7	DeepX	269.44	H8/b066	330.14	+22.5

concurrency settings. The deployment comparison does not hold pipeline organization constant across implementations.

The detector comparison contract specifies original-image $xyxy$ coordinates in pixels, a probability score, and an integer class index. It uses a score threshold of 0.25 and at most 300 detections. Where NMS is required, the policy is class-aware with an IoU threshold of 0.45. The integrated YOLO26 paths retain their graph-defined completion behavior. Table 7 separates these paths from raw-head execution.

All 96 detector configurations have a runtime comparison contract, with the same contract hash within each model across Generic, Native, and Full. The source projection also contains 65 explicit quality comparison objects and 52 explicit performance comparison objects. Every available object matches its runtime counterpart. The completed artifact audit supplies the remaining source bindings. Matching contracts define the required result, while dataset evaluation measures numerical quality.

The public `r20/endpoint_contracts.csv` projection retains the original endpoint checks for 20 central configurations, including all fourteen global winner/reference configurations. These checks confirm required postprocessing and materialization inside timing. The detector projection records the contracts for all 96 configurations and their source hashes. Section D describes the wider 450-configuration audit and the separate checks with different inputs.

The performance protocol uses three independent processes with 1,000 measured tasks each. Available pathway records specify 100 warmup tasks, completed and drained before timing. Timing includes fill, necessary copies, required task completion, and drain. Classification completes after synchronized output materialization and Top1/Top5 processing. Detection follows Table 7. The independent correctness oracle for the 36 Native detector splits runs outside timing. Required detection completion remains inside timing.

Energy runs admit work for 60 s and drain all admitted tasks. The primary metric is the mean of three calibrated

Table 5. Observed repetition minima and maxima for global winners and additional YOLOv7/H8 Full references. Performance and energy use separate runs; these ranges are not confidence intervals. The original Refresh16 YOLO26s/H8 b021 repetitions complete the R22 extract gap.

Deployment	FPS	mJ/image
MobileNetV3-L / DeepX Full	1458–1470	11.98–12.05
MobileNetV3-L / DeepX/b135	1441–1570	12.58–12.59
RegNet-X / DeepX Full	538–541	31.48–31.63
RegNet-X / H10/b123	1103–1386	15.78–15.88
ResNet-50 / DeepX Full	655–659	27.64–27.66
ResNet-50 / H10/b095	1074–1114	29.42–29.50
YOLO11l / DeepX Full	63–63	314.56–315.72
YOLO11l / H10/b120	53–53	537.63–567.69
YOLO26m / TRT (H8)	85–85	399.41–400.41
YOLO26m / H10/b043	110–111	319.55–319.76
YOLO26s / TRT (DeepX)	152–152	186.79–188.63
YOLO26s / H8/b021	187–188	157.56–157.72
YOLOv7 / DeepX Full	83–84	268.19–270.38
YOLOv7 / H8/b066	112–113	329.43–330.50
YOLOv7 / H8 Full	32–32	549.54–550.25
YOLOv7 / TRT (H8)	53–53	596.63–599.40

E_r/N_r values. Total joules, durations, counts, and individual repeats remain in the source records. Repeated prepared inputs define a throughput microbenchmark. Reciprocal throughput is not per-request latency.

D Artifact Audit and Eligibility

The audit connects dataset quality, compiled artifacts, prepared inputs, completed-task endpoints, performance repetitions, and energy records for all 450 configurations. Configuration names alone do not establish a match. All 407 configurations retained by the 5% quality rule have the required bindings, including the fourteen global winners. The other 43 fail numerical quality.

The 75 DeepX Native configurations already used the uniform A4 policy. Their 225 performance processes and 225 energy repetitions preceded the audit. Each energy configuration has three calibrated 60 s repetitions under the final acquisition protocol. Earlier screening labels reflected a previous admission and quality classification. Six YOLO26 split aggregates retained partial status after files moved, despite three valid raw repetitions and matching artifact hashes. Seventeen quality-receipt version discrepancies were resolved against the evaluation files, their hashes, execution reports, and runtime artifact chain. Original records remain unchanged, and corrected inclusion decisions are separate derived data.

Five Hailo Full configurations have contemporaneous artifact hashes before and after all three repetitions. The other nine have composite production records connecting each exact performance report to its HEF, graph, and preparation. These records support the Full references without reconstructing every historical software condition.

Distinct-input checks cover the seven DeepX Fulls and MobileNet/DeepX b135. For b135, all 32 serial references and asynchronous outputs match exactly at the boundary and logits. These configuration-specific checks do not establish arbitrary input/output association across all 450 configurations or add new quality, throughput, or energy measurements.

E Selection Methods and Calibration

Within each model/setup group, Generic selects the cut with the highest measured Generic throughput. Its loss compares that cut's Native throughput with the best measured Native throughput in the same group. Configuration identity resolves ties, and groups receive equal weight. The mean one-choice loss is 15.08% for the 187 quality-eligible pairs, compared with 15.16% for all 204 pairs. The repository retains the complete unfiltered group results.

E.1 Device-specific Generic calibration

The fixed calibration uses 151 eligible classifier pairs in nine model/setup groups. For each setup, one complete architecture is held out and the other two supply training data. The formula and penalty were fixed before fitting. For candidate i, the model predicts

$$\widehat{\log R_{N,i}} = \log R_{G,i} + \beta_0 + \beta_1 z(x_i) + \beta_2 z\big(\log(1+b_i)\big), \quad (1)$$

where x_i is the prefix share of graph-estimated computation and b_i is logical boundary size in bytes. Training minimizes

$$\frac{1}{n}\sum_i (y_i - \widehat{y}_i)^2 + \beta_1^2 + \beta_2^2, \qquad y_i = \log(R_{N,i}/R_{G,i}). \quad (2)$$

The intercept is unpenalized and the ridge penalty is one. Standardization uses training-row means and population standard deviations. Constant features receive scale one and standardized value zero. There is no feature selection, hyperparameter search, or random cut-level train/test split.

Verified source graphs supply the features. Bytes ranks by increasing boundary size, Generic by measured Generic throughput, and calibration by predicted Native throughput. Configuration identity breaks ties. Each shortlist retains its fastest measured Native member, with loss relative to the eligible Native optimum in that group. The nine groups receive equal weight.

Calibration reduces mean one-choice loss from 32.80% to 28.32%, and five-choice loss from 20.45% to 15.43%. Bytes

Table 6. Recorded setup and workload metadata. The YOLOv7 platform observations describe its setup comparison. Other deployments retain their own recorded settings.

Item	Recorded setting	Evidence scope
Performance workload	Three repetitions and 1,000 completed tasks per repetition	204 split and 42 Full configurations. YOLOv7 combines 27 new split and 18 preserved Full repetitions
Batch and Generic queues	Batch 1. two workers. queue depth 2. 100 warmups. one exactly prepared feed repeated	Nine targeted pathway records, not all 204 splits
Native queues	Queue depth 3. 100 warmups. H10 paths additionally record 8 prefix requests in flight	The same nine targeted pathway records. runtime organization remains case-specific
Nominal spatial inputs	ImageNet validation 224×224 and COCO 2017 validation 640×640	Dataset identities and available preprocessing contracts, described below
Orin configuration	NX 16 GB. L4T 36.4.7. matching active `MAXN_SUPER` definition	YOLOv7 setup comparison
Clock and link observations	CPU 1984 MHz. GPU 1173 MHz. no observed OC-counter increments. accelerator PCIe snapshots 8.0 GT/s $\times 4$	Approximately 1-Hz YOLOv7 load samples and separate link snapshots. EMC unobserved
TensorRT Full build	Same model, FP16 option, static FLOAT input/output shapes and workspace setting	YOLOv7 Full comparison. serialized engines and internal tactics need not be identical
DC energy boundary	Calibrated full-system input energy. no idle subtraction in the primary comparison	231 cases / 693 repetitions over six models. calibration detail in Section H

Table 7. Required task outputs and completion work. Generic, Native, and Full use the same model-specific comparison contract. The compiled graph determines whether decoding and suppression occur in the model or in host code. Required output processing and materialization precede task completion.

Model or output path	Completed result	Required processing
MobileNet, RegNet, ResNet	Synchronized classifier outputs and ordered Top1/Top5	Output materialization and Top1/Top5 processing.
YOLO11l and YOLOv7	Original-image boxes, probability scores, and class indices	Model-specific decoding and class-aware NMS where not already completed by the model, followed by materialization.
YOLO26m/s, raw-head paths	The same detection record format	Decode raw heads and apply the required NMS. This includes Hailo Full paths.
YOLO26m/s, integrated outputs	The same detection record format	Filter and normalize integrated detections, undo letterboxing, and materialize outputs. No additional NMS pass is applied to valid integrated outputs.

reaches 8.86% at five choices. The average calibration improvement comes from DeepX. Calibration is worse than Generic on average across H10 classifiers at each evaluated budget ($k = 1, 3, 5$). On H8, five-choice mean loss rises from zero to 1.825%. The repository retains the complete per-group results.

This calibration supplies no detector predictions. The held-out architecture contributes no training rows, but all three architectures were known when the study was designed. Across overlapping folds, 151 distinct Native and 151 distinct Generic configurations are reused. The budget counts only Native trials in the held-out group. Training, Generic acquisition, compilation, and quality evaluation are excluded, and total search-cost savings remain unmeasured.

E.2 Shared graph-score weights

The current-data score comparison includes all seven models and all 187 quality-admissible Native/Generic pairs. The 151 classifier pairs form nine model/setup groups, and the 36 detector pairs form twelve. Each group has equal weight at every budget. All methods use the same measured candidates and the same Native optimum. A group with n candidates tests $\min(k, n)$ cuts, so the three-cut detector groups remain in the average with zero loss from $k = 3$ onward.

Current graph features. Graph and boundary hashes bind every pair to its source ONNX graph. The full workflow candidate sets contain 138, 133, 120, 615, 416, 382, and 307 cuts for MobileNetV3-L, RegNet-X, ResNet-50, YOLO11l, YOLO26m, YOLO26s, and YOLOv7. These sets define normalization ranges, not measured performance

Table 8. Additional analytic operation rules. L is the reduction-vector length. TopK selects K values from length N. Counts are per vector except for elementwise Mod.

Operation	Count
Stable Softmax	$6L - 3$
ReduceMax	$2(L - 1)$
Integer Mod	One per output element
TopK reference	$3\sum_{j=0}^{K-1}(N - j - 1)$

coverage. The logical size of all crossing tensors supplies communication. The score is

$$s(c) = w_{\text{comm}}\widetilde{C}(c) + w_{\text{imb}}I(c) + w_{\text{ten}}\widetilde{T}(c), \qquad (3)$$

where $C = \log_{10}(1 + b)$ for boundary bytes b, $I = |F_L - F_R|/(F_L + F_R)$, and $T = \max(n_{\text{tensors}} - 1, 0)$. Tildes denote min/max normalization over the full workflow candidate set before measurement or quality filtering. A constant feature becomes zero. The imbalance term is used directly, as in the printed ARCS formula (Mika et al., 2026).

Operation-count convention. The analytic count treats one MAC, scalar arithmetic primitive, comparison, or selection as one operation. Structural data moves are excluded. These are graph-cost proxies, not measured processor instructions or stage times. Existing complete counts for the three classifiers and YOLOv7 are preserved. A documented extension supplies counts for fifteen previously unsupported Softmax, ReduceMax, Mod, and TopK nodes in the other three models. No missing count is replaced by zero.

Softmax includes finding and subtracting the maximum, exponentiation, summation, and division. ReduceMax uses one comparison and one selection per step. Mod counts one scalar remainder primitive. TopK uses a declared repeated-maximum reference algorithm without replacement. Each scanned candidate requires a comparison and separate value and index selections. It scans original indices in ascending order and retains the earlier index on ties. This rule does not assert that a runtime uses that algorithm.

The captured graphs use ONNX opset 18. Shapes uniquely recover the omitted ReduceMax axis constants and TopK values. The reduction $[1, 8400, 80] \to [1, 8400]$ with dimensions removed determines axis 2, and the value and index outputs $[1, 300]$ determine $K = 300$. The exporter rejects ambiguous deductions. The added counts total 10,795,200 operations for YOLO11l and 37,887,000 each for YOLO26m and YOLO26s. Multiplying only these new contributions by 0.1 or 10 changes none of the complete rankings under the reported default, joint, classifier-trained, or held-out weights. That sensitivity check keeps weights fixed.

Table 9. Weights fitted with one complete model held out. Each fit uses the other six models on all three devices. The communication weight fixes the scale. The tensor weight remains fixed because all measured cuts have zero tensor penalty.

Held-out model	w_{comm}	w_{imb}	w_{ten}
MobileNetV3-L	1	0.569343	0.2
RegNet-X	1	0.157993	0.2
ResNet-50	1	0.025085	0.2
YOLO11l	1	0.157993	0.2
YOLO26m	1	0.157993	0.2
YOLO26s	1	0.157993	0.2
YOLOv7	1	0.157993	0.2

Fitting and held-out evaluation. Default Weighted uses $(w_{\text{comm}}, w_{\text{imb}}, w_{\text{ten}}) = (1, 3, 0.2)$. Every measured boundary has one crossing tensor, so its tensor penalty is zero. The tensor weight cannot be estimated from these cases and remains fixed. We search all nonnegative communication-to-imbalance ratios through

$$s_\alpha(c) = (1 - \alpha)\widetilde{C}(c) + \alpha I(c), \qquad 0 \le \alpha \le 1. \qquad (4)$$

The endpoints include pure boundary bytes and pure imbalance. Configuration identity resolves ties, including ties under the pure-byte endpoint.

The objective is mean percentage Native throughput loss at $k = 1, 3, 5$, with equal weights across groups and budgets. An exhaustive one-dimensional search evaluates every interval where the ranking is constant and every boundary between intervals. The midpoint of the widest optimal interval supplies a representative weight. A smaller α resolves equal-width ties. An independent 1,001-point grid checks the minimum. For $\alpha < 1$, the reported weights are $w_{\text{comm}} = 1$ and $w_{\text{imb}} = \alpha/(1 - \alpha)$.

The joint fit uses all 21 groups and gives $(1, 0.15799267, 0.2)$. A fit using only the nine classifier groups gives the same weights. The latter is a transfer test for all four detectors, but its classifier results remain training-data results. In the separate model-held-out evaluation, each model is excluded across all three devices. The remaining six models supply eighteen training groups, and the excluded model supplies three test groups. The displayed held-out curve combines only these test results. The normalized graph features of a held-out model use its graph structure, without its Native throughput.

Across overlapping folds, this analysis reuses 187 distinct Native configurations. Budgets count Native trials in the evaluated group and exclude fitting measurements, Generic acquisition, compilation, and quality evaluation. They do not establish total search-cost savings.

The public export contains the features and source bindings, complete rankings and per-group curves, fold memberships

and weights, and sensitivity results. The figure and budget table are generated from these data. The separate Generic calibration supplies no detector predictions and is excluded from the 21-group curve.

F Controlled Scheduling Experiments

The experiments use RegNet/H10 b052 and b123, and ResNet/H10 b002 and b095. Both pairs were selected for a known Generic/Native ranking reversal. Each experiment has 40 fresh processes in five complete randomized blocks. All 80 processes were admitted without failures, exclusions, or replacements. The earlier 20-process Native-only ResNet study and eighteen H10 pilots remain separate in the repository and do not enter these estimates.

Intervention and runtime chains. Each block contains both cuts under Generic and Native, each with productive scheduling and globally at most one outstanding task. The eight cells run sequentially in randomized order, each in a fresh worker process. Productive Generic uses two workers and queue limit two. Productive Native uses eight asynchronous Hailo slots and FIFO limit three. In the single-task condition, a global completion gate holds admission ownership through output materialization, synchronized TensorRT execution, and Top1/Top5 processing. Native also uses one asynchronous slot. The global gate is needed because reducing the slot count alone would not enforce this completed-task boundary.

Each runtime retains its backend, SDK, handoff, conversion, and materialization paths. The experiment changes scheduling within these chains without independently switching their other differences.

Artifacts, timing, and device state. The experiments retain the deployment cases' compiled HEFs, TensorRT engines, precision settings, and prepared input bytes. They use frozen current runtime sources. Each process warms up for 30 s, drains, admits measured work for 60 s, and drains again. Throughput divides completed measured tasks by the full makespan, including fill, materialized outputs, Top1/Top5 processing, and drain. ResNet makespans range from 60.00138 to 60.02365 s. No energy measurements were added.

Before and after each run, checks cover clocks, power profile, affinity, temperature, competing work, and file bindings. Settings are MAXN_SUPER, 1.984 GHz CPU, 1.173 GHz GPU, and CPU affinity 0–7. Admission requires a starting CPU temperature of 40–65 °C, at most 2 °C drift over 2 s, and no active throttle alert. The protocol allows at least 5 s rest and at most 120 s waiting for admission. End temperatures are descriptive, with no post-hoc exclusions. ResNet reaches up to 66 °C.

Functional checks. Sixteen separate H10 pilot processes each exercise 32 mixed requests from two distinguishable inputs. Outputs match byte for byte across runtime chains, schedules, and saved functional references. The full ResNet replication reuses its eight checks (256 requests), with sources, artifacts, inputs, and paths unchanged. These checks do not enter throughput estimates or replace dataset-quality evaluation. The single-task constraint is supported by the pinned completion-gate ordering, available queue and inflight counters, and functional checks. A direct measured global high-water counter is unavailable.

F.1 Contrasts and uncertainty

Let $r_{v,s,b}$ be the later-cut/earlier-cut throughput ratio in block b, for runtime $v \in \{G, N\}$ and schedule $s \in \{P, S\}$. RegNet uses b123/b052 and ResNet uses b095/b002. The scheduling contrast is $r_{v,P,b}/r_{v,S,b}$, and the interaction is

$$I_b = \frac{r_{N,P,b}/r_{N,S,b}}{r_{G,P,b}/r_{G,S,b}}. \tag{5}$$

Contrasts are averaged on the log scale over five blocks and exponentiated. For log contrast ℓ_b, the nominal 95% interval is

$$\exp\left(\bar{\ell} \pm 2.776445\frac{s_\ell}{\sqrt{5}}\right), \tag{6}$$

where s_ℓ is the sample standard deviation and $2.776445 = t_{0.975,4}$. Blocks are the replication units. The intervals have no simultaneous coverage correction.

The RegNet protocol allowed a separate five-block extension if any log interval for productive Generic, productive Native, or the interaction included zero. None did, so no extension ran. This conditional design does not establish sequential or simultaneous 95% coverage. RegNet's mathematical rules preceded measurement, while its additional implementation-hash record was created after the first process began and before the first complete block. ResNet's protocol and analyzer were frozen before its first process, with five fixed blocks and no outcome-dependent extension.

The main paper reports the aggregate contrasts. Table 10 retains all blockwise values. Under productive scheduling, Generic prefers the earlier cut and Native the later cut in both experiments. Under single-task execution, both prefer b123 for RegNet and b002 for ResNet. These selected cases establish scheduling-dependent ordering within the measured chains, not its prevalence or an isolated copy or conversion cause.

Supporting Native counters. For productive RegNet b052 and b123, median serial-consumer intervals are 2.218

Table 10. Blockwise contrasts in the two full experiments. Each block contains eight independent processes measured in randomized order. GP/NP are Generic/Native productive ratios; GS/NS are single-task ratios. Scheduling contrasts divide productive by single-task; interaction divides Native scheduling by Generic scheduling.

Series	Block	GP	NP	GS	NS	G sched.	N sched.	Interaction
RegNet	1	0.65765	3.36887	1.02336	1.03003	0.64264	3.27064	5.08936
RegNet	2	0.66320	3.38481	1.02897	1.03296	0.64452	3.27681	5.08408
RegNet	3	0.65670	3.37013	1.03603	1.02491	0.63386	3.28822	5.18760
RegNet	4	0.66496	3.35211	1.02540	1.03995	0.64849	3.22334	4.97053
RegNet	5	0.65928	3.38085	1.03061	1.01393	0.63970	3.33440	5.21247
ResNet full	1	0.60584	2.01615	0.83901	0.83102	0.72209	2.42613	3.35988
ResNet full	2	0.60686	2.06184	0.83123	0.82767	0.73008	2.49113	3.41214
ResNet full	3	0.60280	2.07828	0.83492	0.82749	0.72198	2.51155	3.47871
ResNet full	4	0.60805	2.07796	0.84567	0.82412	0.71901	2.52144	3.50682
ResNet full	5	0.60502	2.06515	0.82861	0.83464	0.73017	2.47431	3.38869

and 0.613 ms, accounting for 98.98% and 92.29% of measured duration. Synchronized TensorRT work, including H2D, execution, D2H, and synchronization, falls from 2.020 to 0.451 ms. FIFO put blocking falls from 1.675 to 0.021 ms. These counters support a stronger consumer bottleneck at b052. Pipeline intervals overlap and must not be added as independent costs. Corresponding Generic stage counters are unavailable.

The independent stage-cost experiment in Section G uses RegNet b059/b123 and tests a composition rule. It is separate from the scheduling experiment's b052/b123 intervention.

G Independent Stage-Cost Experiment

The experiment uses ResNet-50/DeepX b045 and b002, and RegNet-X/H10 b059 and b123. Each case has three blocks with separate producer (P), consumer (C), and pipeline (PC) processes: 36 performance processes and twelve stage-composition comparisons. Four serial-reference processes and twelve smokes are additional preparation.

Each process uses batch one, 100 drained warmups, and 10,000 measured completions. The fixed middle-window rate is $8000/(t_{9000} - t_{1000})$, with times in seconds and one-based completion order. Total rates retain fill and drain. The two halves of this window support the predefined 5% drift check. No interval is chosen after measurement.

P ends when a usable host-side boundary tensor is available, including output mapping and ownership transfer. DeepX uses a synchronous producer here, while H10 uses its stored eight-request producer. C rotates four genuine saved boundary buffers and includes host staging, H2D, TensorRT execution, D2H/synchronization, and ordered Top1/Top5. PC uses the same functions with a FIFO of depth three and timed per-task copies. Replayed buffers and fresh PC payloads can differ in cache and allocation behavior.

Within each block, $\widehat{F}_{PC} = \min(F_P, F_C)$ is formed before reading the PC report. Signed error is $100(\widehat{F}_{PC}/F_{PC} - 1)$. The cycle residual is $1000/F_{PC} - \max(1000/F_P, 1000/F_C)$ in milliseconds. These are effective intervals, not request latencies. Errors are calculated per block before aggregation.

All twelve predictions are optimistic. The largest individual error is 8.80%, below the predefined practical 10% band. Both within-model orderings agree in every block. The four-case summary appears in the main paper. These measured capacities test a composition rule, not a low-cost predictor for unseen cuts.

Each recipe repeats one prepared input. Integer parity is exact. Floating-point checks use $\mathrm{rtol} = 10^{-4}$ and $\mathrm{atol} = 10^{-5}$, with matching ordered Top1/Top5. The recorded settings are MAXN_SUPER, 1.984 GHz CPU and 1.173 GHz GPU clocks. Clock, affinity, temperature, and missing-EMC observations remain attached to the reports. Full block results are in `COMPOSITION_COMPARISON.csv`. These measurements use their own runtime and timing window rather than the deployment or scheduling rates.

H Energy Calibration and Measurement Windows

Energy covers DC input to the host and attached accelerator, excluding upstream AC/DC losses. The setup uses an input shunt and INA225 acquisition. Stored calibration gains are 0.913634 for H8, 0.896397 for H10, and 0.903508 for DeepX, from the original load/idle fits. A common multiplicative gain cancels in a same-setup ratio, but generally not across setups. Load-dependent errors need not cancel.

H10 retains calibration warnings for a 2.026% point-factor spread and a gain below a configured 0.90 bound. Calibration captures use 2 kS/s and 30 s intervals. These settings do not specify every experiment's acquisition or a complete uncertainty budget. Separate command-window records es-

tablish 2 kS/s acquisition for all 225 energy repetitions of the 75 DeepX Native configurations. Their source hashes and calibration bindings remain in the evidence bundle linked in Section I.

Longer-window comparison. The separate six-model study contains 693 repetitions with completion windows of 61.28–63.10 s. It uses N_r/T_r from those windows, rather than independent performance medians. Sixteen of eighteen quality-filtered model/setup groups retain the same rate winner between the shorter performance and longer energy-window measurements. YOLO11l/H10 changes from b120 to b062, and YOLO26m/DeepX from b043 to b038.

This study uses earlier runtime paths and does not include the current DeepX Full implementation. It therefore does not repeat the main global-winner comparison over longer windows. For example, its YOLO11l b062 achieves 37 FPS at 641.55 mJ/image, against 36 FPS and 707.36 mJ/image for its fastest eligible Full. In the current deployment data, b062 reaches 51 FPS at 558.64 mJ/image, while DeepX Full reaches 63 FPS at 315.23 mJ/image. These windows do not establish indefinite thermal stability.

I Software and Reproduction

The ONNX-Splitpoint-Tool generates and compiles subgraphs and coordinates hardware measurements. It was developed with large-language-model coding assistance under author-defined requirements. Regression tests, numerical checks, and hardware runs evaluated the implementations. The authors remain responsible for the software, experiments, and interpretation. Programming productivity was not measured.

Offline reproduction reconstructs tables, rankings, and figures from recorded inputs. It requires neither model artifacts nor accelerator hardware and does not acquire new measurements. The repository's MLSys entry page links the publication outputs, evidence, and required Python environment (Mika, 2026).

From the repository root, run:

```
python -I -B \
scripts/split-inference/reproduce.py \
--output /tmp/edge-ai-reproduced \
--publication mlsys-r30 --verify
```

Use an empty output directory. The `mlsys-r30` profile generates the current tables and figures with shared typography and colors, including both YOLOv7 layouts. Outputs appear under `publications/mlsys-r30/` in that directory. Adding `--numeric-only` omits figure rendering. The asset manifest records inputs and generators. Within this LaTeX package, `make all` compiles both documents from the supplied assets.

Supporting records. The repository retains the following reproducible evidence:

- Weighted analysis: bound graph inputs, operation rules, all rankings and curves, fold memberships, fitted weights, and sensitivity checks.
- Audit and calibration: all 450 inclusion decisions, flag resolutions, classifier calibration, and original RegNet scheduling records.
- Full ResNet replication: forty fresh process records and its fixed protocol. Evidence manifests preserve original archive and member hashes for both scheduling studies.
- Stage-cost study: all 36 process reports, twelve block comparisons, and the independent evaluator.
- Earlier selection study: the separate long-window analysis, vendor quality and feasibility results, and their original runtime references.

Earlier analyses. The complete earlier RQ2 study retains the five fixed graph rules and their original Native measurements. The frozen `mlsys-r22` profile preserves the earlier budget, calibration, and Native-only ResNet tables. These analyses remain separate from current deployment results. The R31 supplement and source archive preserve the previous presentation, including compiler-interface diagnostics whose original raw probes are not part of the public compact inputs.

References

Mika, K. Edge AI performance–energy results. Versioned research artifact, 2026. URL https://github.com/Keff789/edge-ai-performance-energy-results/tree/eba5baac2990baffa71c269fd1402566ea8f1198. Source snapshot: commit eba5baa.

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